Ageing Research Reviews 119 (2026) 103181

Contents lists available at ScienceDirect

Ageing Research Reviews

journal homepage: www.elsevier.com/locate/arr

Review article

The Hallmarks of aging: Paradigms and scientific progress

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ARTICLE INFO

Keywords:
Aging
Damage
Hallmarks
Geroscience
Paradigm
Scientific progress

ABSTRACT

The Hallmarks of Aging framework has become a widely accepted schema within geroscience, organizing diverse mechanisms of cellular and molecular decline. This article examines whether the Hallmarks function as a scientific paradigm and, beyond that, whether they have enabled genuine scientific progress—an angle largely absents from current literature. Drawing on models from the philosophy of science, the article evaluates the framework's status and impact from multiple perspectives. From Kuhn, the Hallmarks act as exemplars within the damage paradigm, guiding "normal science" without instigating revolutionary change. From Lakatos, they form part of a protective belt that stabilizes a research programme rather than producing theoretical innovation. From Laudan, they help solve empirical problems but leave conceptual issues unresolved. From Kitcher, their success lies in rhetorical and institutional unification more than inferential structure. From Galison, they are best understood as an instrumental "trading zone" that enables cross-disciplinary coordination. In sum, while the Hallmarks have brought coherence and methodological infrastructure to aging research, they may also constrain conceptual imagination—raising critical questions about the field's long-term epistemic vitality.

1. Introduction

Aging matters. Many lives are at stake (Bostrom, 2005; de Grey, 2005; Gems, 2003). We are facing one of the greatest challenges of our time (Farrelly, 2008, 2010, 2022). Within this context, geroscience develops as a field that identifies aging as the primary risk factor for most age-related diseases and, consequently, as a central target for biomedical interventions (Kennedy et al., 2014). As a result, we have reached a point where it is no longer enough for us to simply understand it in theory; given its negative effects, we now aim to take action to address it (García-Barranquero et al., 2024a; García-Barranquero et al., 2024b).

Driven by new funding streams, high-profile initiatives, and the increasing convergence of biotech innovations, this momentum promotes increasingly ambitious narratives centered on the possibility of slowing—or even arresting—aging (de Grey and Rae, 2007; Sinclair and LaPlante, 2019). At the same time, it generates expanding institutional support and strategic research agendas aim at positioning geroscience at the forefront of biomedical priorities (Kaeberlein, 2017; Longo et al., 2015).

On the other hand, this shared sense of progress lies a less visible but equally serious crisis within the field (Poganik and Gladyshev, 2023). Newly international surveys indicate that scientists are divided on basic questions, including what aging is, when it starts, and whether it should be thought of as programmed, stochastic, or possibly reversible process (Cohen et al., 2020a; Gladyshev et al., 2024).

However, alongside these high expectations surrounding the promise of curing aging, and despite the lack of theoretical consensus within the field, one framework has achieved remarkable visibility and influence: The Hallmarks of Aging. This was initially offered as a synthetic review of central molecular and cellular mechanisms in aging (López-Otín et al., 2013). This structure is now widely used as a general organizing schema that serves as a quasi-default framework for all practitioners in the field. The 2013 model was comprised of nine hallmarks: genomic instability, telomere attrition, epigenetic alterations, loss of proteostasis, deregulated nutrient sensing, mitochondrial dysfunction, cellular senescence, stem-cell exhaustion, and altered intercellular communication. They were put into three categories: *primary hallmarks*, "the initial origins of molecular damage"; *antagonistic hallmarks*, "adaptive and initially beneficial responses that become progressively detrimental as damage accumulates"; and *integrative hallmarks*, "which emerge when the combined effects of primary damage and antagonistic response exceed the

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https://doi.org/10.1016/j.arr.2026.103181
Received 19 December 2025; Received in revised form 20 April 2026; Accepted 14 May 2026
Available online 14 May 2026

organism's recovery rate and its homeostasis in tissue, with consequent functional impairment" (López-Otín et al., 2013: 1208–9).

A decade later, the Hallmarks was adapted from results as it received popular support from the academic literature and serves as a guideline for the orderly organization of experimental findings (López-Otín et al., 2023). This revised edition used knowledge of more than 10 years of research using the Hallmarks as a guiding framework and added three more hallmarks—chronic inflammation, extracellular matrix dysregulation, and microbiome alterations—bringing the total to 12.

The impact of the Hallmarks can be assessed along two complementary dimensions. Firstly, the current citation counts reflect the contribution of this framework. The original 2013 article has about 18,800 citations, and the 2023 update has around 6000. Compared to other review articles, nearly none have similar data effects. The most recent example is a survey of the 100 most cited publications in aging research that found that the Hallmarks was one of the most impactful frameworks in geroscience (Haroon et al., 2022).

Second, conceptually, the influence of the Hallmarks is shown through the higher number of publications that use the review. Some articles expand the definition with newly defined hallmarks, such as chronic inflammation (Campisi et al., 2019) or dysregulation of the extracellular matrix (Schmauck-Medina et al., 2022). Other articles re-evaluate the connections between the established hallmarks (Kroemer et al., 2025), and many move the model on to adjacent conceptual programs, like the hallmarks of health (López-Otín and Kroemer, 2021), or the hallmarks of rejuvenation (Nunkoo et al., 2024). The proliferation of derivative frameworks proves that the Hallmarks model also constitutes not only a heuristic tool, but also rhetorically and organizationally as well (Tartiere et al., 2024).

The rapid integration of the Hallmarks as a central reference point in geroscience warrants in-depth analysis (Kaeberlein, 2025). Despite its major influence since 2013, discussions of the framework has focused largely on empirical developments rather than its methodological and theoretical foundations. Even the 2023 update, while synthesizing a decade of experimental progress, offers no reflection on the conceptual assumptions and limits of the model. However, the existing literature has discerned three replies, which reveal the fundamental tensions at the heart of the Hallmarks. The first challenges the notion that aging constitutes a biologically unified process governed by a conserved set of mechanisms across taxa (Cohen, 2018). The second concerns the methodological deployment of those Hallmarks in experimental and translational studies, especially the reliance on lifespan extension as a primary operational endpoint (Keshavarz et al., 2023). Instead, we will focus on a third and structurally more focused critique advanced by Gems and de Magalhães (2021). They argue that, in spite of its integrative appeal, the framework lacks a principled causal architecture of its own. To put it more simply, they address the confusion associated with distinguishing between primary drivers of aging and downstream or compensatory processes, the lack of a defined hierarchy among the hallmarks, and the permissive structure that allows additional hallmarks to be incorporated without putting the theoretical core at risk.

This latter concern drives our article. Gems and de Magalhães (2021) especially ask if the Hallmarks have become a core organizing framework for aging research. Importantly, this concern does not arise away from the field. In the geroscience literature, similar philosophical words like these have already begun to appear in passing, as Kuhnian paradigms or Lakatosian research programmes are sometimes called on to indicate a lack of consensus (Cohen et al., 2020a). This indicates that exploring the theoretical status of the Hallmarks is not an external philosophical directive, but is rather a development of questions that were already growing out of the geroscience literature. Meanwhile, the foundation issues often take a back seat to empirical discovery, and institutional incentives promote experimental novelty versus theoretical explication (see DiFrisco and Orzack, 2026). Consequently, even the most influential theoretical frameworks can serve as a basis for the conduct of research without a thorough exploration of their epistemological pillars.

Methodologically, our approach is primarily theoretical. We build on the framework proposed by Laplane et al. (2019) highlight four main contributions of philosophical analysis to the life sciences: clarifying scientific concepts; critiquing background assumptions; reconstructing explanatory frameworks; and articulating connecting levels of explanations.[1] Although our work is conceptually-oriented, we will selectively rely on a range of empirical markers, including bibliometric trends and changes in the research scenery, to substantiate our arguments as well (complementarily, Pradeu et al., 2024). We should also bear in mind that the Hallmarks did not develop in isolation. Instead, it is an outgrowth of previous efforts to organize the biology of aging, and it coexists with other integrative theories that have coexisted and competed for intellectual space within recent years, something that seems a little forgotten in the literature (de Grey, 2023; García-Barranquero and Pérez-González, 2025). The Hallmarks must therefore be considered in this wider context to understand both their impact and their constraints. In this context, we will first put the Hallmarks through the perspective of Kuhn's (1970) conception of scientific paradigms to explore whether the framework has reconfigured the geroscience's field. We will then utilize Lakatos' (1978) account of research programmes, less rigid then Kuhnian theory. This analysis leads naturally to another question: whether the Hallmarks have contributed to scientific progress in geroscience or not. We will attempt to answer by referring to an account of philosophical advancements such as the efficacy of Laudan's (1978) problem-solving approach, Kitcher's (1993) theory of explanatory unity, and Galison's (1997) focus on methodological innovations.[2]

2. Paradigms and research programmes

2.1. *From evolutionary theory to molecular mechanisms*

Nothing comes from nothing, and the Hallmarks are no exception. We will briefly present a historical overview. Scientific explanations for aging originated from the evolutionary theory—formulations of Medawar (1952), Williams (1957), and Hamilton (1966). The central thesis shared by these authors is that aging results from inevitable weakening of natural selection as an organism grows older. They argued that living beings age because of evolutionary pressures and a preference for early reproduction over long-term maintenance. These evolutionary narratives did not simply explain the cause of aging but indicated how it might manifest itself, most notably in Williams' idea of antagonistic pleiotropy that explicitly connected genetic processes to age-dependent decline.

Following this tradition, Kirwood's (1977) disposable soma theory detailed the link between ultimate evolutionary explanations and proximate physiological mechanisms. Kirkwood reframed aging as the cumulative cost of underinvestment in repair processes by suggesting that organisms allocate finite resources across reproduction and the maintenance of the somatic system. So, in this sense, his contribution was at no point a substitute for evolutionary explanations, but was a technical translation of the evolutionary model into a mechanism.

[1] This is also the case in domains such as social evolution, where a number of philosophical contributions (e.g. Birch, 2017a, 2017b; Birch and Okasha, 2015; Okasha, 2006, 2011) have played a significant role in shaping scientific debates concerning the mathematical equivalence, causal interpretation, and conceptual foundations of inclusive fitness theory and multilevel selection theory.

[2] Of course, the Hallmarks may also be assessed from alternative perspectives. For instance, they could be examined through the lens of prominent philosophical frameworks on causality (e.g. the manipulationist approach; Woodward, 2003), with a view to evaluating their capacity to provide causal evidence; or they could be treated as case studies for debates in scientific realism, particularly regarding their potential role in guiding investigations that track what scientists take to be true. While these approaches are undoubtedly relevant, we take them to be orthogonal to the primary aims of the present article.

Concurrent with these theoretical advances, experimental evidence had already demonstrated that aging could be assessed at the cellular level. Hayflick and Moorhead's (1961) theory revealed that human somatic cells had a finite replicative capacity was one early empirical way to demonstrate the fact that limits related to aging could be observed, measured, and manipulated experimentally in vitro. These insights were further elaborated at the molecular level in subsequent work. Holliday (1997), for example, highlighted DNA repair, genome stability, and epigenetic regulation as proximate mechanisms of physiological decline. These contributions together represented a definitive direction of explanatory focus—away from the ultimate evolutionary sources of change towards a more specific set of mechanisms—and contributed significantly to aging being recognized as a biomedical object, and, in principle, one that can be modified.

2.2. *The damage framework*

From our perspective, these ideas can be interpreted as the damage-and-maintenance paradigm (Gems and Partridge, 2013: 626). Taking it a step further, Holliday's article (1998) was particularly notable in this regard, describing aging as a progressive degradation of the body's multiple maintenance systems, including DNA repair, proteostasis, antioxidant defenses, epigenetic stability, immune surveillance, and structural integrity, due to their inherent imperfection. In this vein, aging didn't originate in one causal root but rather because of a long series of broken processes lying throughout the organism. His model described why aging arises, why it is universal, and how it comes from the physiological limitations, a byproduct of how we evolved. In doing so, it opened up a horizon in which aging could be approached not only as an object for understanding, but also as something that might actually be targeted through intervention (see Section 1).

The new millennium developed this ontology even further turning an explanatory model into proposals in which biomedical intervention became paramount, if not an explicit aim. The SENS programme (de Grey et al., 2002) reinterpreted the decline described by Holliday as concrete classes of alterations that could, theoretically, be technically repaired—reinforcing the multicausality of deterioration into seven actionable categories. (see also, de Grey, 2004, 2008). Simultaneously, the Hallmarks of Cancer (Hanahan and Weinberg, 2000, 2011) established a conceptual and visual approach in which intricate oncological phenomena were cohesively distilled into a common language repertoire, providing a unifying story for a fragmented community. The Hallmarks, which sprang directly from that tradition, took Holliday's multicausal schema and structured it as a hierarchical taxonomic map of mechanisms (*primary*, *antagonistic*, *integrative*) that described the course from those initial molecular lesions to systemic physiological impairment (López-Otín et al., 2013). Accordingly, while SENS constituted an intervention-oriented operationalization of the damage-and-maintenance paradigm, the Hallmarks constructed a conceptual map to support its implementation, expansion, and future treatment trajectory (García-Barranquero and Pérez-González, 2025).

The Hallmarks characterized aging by 12 processes that encapsulated the major facets of the damage-and-maintenance paradigm. They were attractive for their visual symmetry and taxonomic order, making the biology of aging legible to all disciplines and across audiences. As with the Hallmarks of Cancer, they granted disciplinary maturity: a unified image that suggested the field had reached a coherence comparable to a "normal science". In light of these factors, we believe that now is the time to think about Hallmarks more seriously (see also, Huang et al., 2025).

2.3. *Hallmarks as Kunhian exemplars*

Building on the earlier discussion, the reply of Gems and de Magalhães (2021) provides a crucial entry point for assessing the Hallmarks' status. Their critique goes beyond suggesting the framework is unexplanatory or dismissible; rather, they challenge the prevailing assumption that the Hallmarks provide a robust theoretical account of aging. Their analysis concludes that the framework serves mainly as a mechanism for classifying and heuristics: it aggregates a disparate set of age-related processes, without (1) the ability to rank cause and effect; (2) distinguishing between drivers and downstream impacts of age; and (3) addressing fundamental questions about why aging occurs. In this regard, they describe the Hallmarks as a "pseudo-paradigm"—not because they are empirically unfounded, but because they are ascribed a status greater than their direct use in the field (Gems and de Magalhães, 2021: 6).

Instead of dismissing this diagnosis, we suggest that the Hallmarks be reinterpreted in relation to a more precise description of scientific organization. To this effect, we use Kuhn's (1970) concept of scientific development as the sequence of alternating stages: pre-paradigmatic diversity, paradigmatic consolidation, and revolutionary transition. In the pre-paradigmatic phase, other groups have incompatible assumptions to work through the conflicts about what constitutes an acceptable problem, methodology, or ontology. Progress is stymied by such fragmentation, because there is no common standard of evaluation or common language. According to Kuhn, a paradigm emerges when one framework is able to reconcile theory, instrumentation, exemplars, and professional practice—establishing a shared worldview that determines what kinds of questions are meaningful, what counts as evidence, and what types of abnormalities contribute to knowledge. And once that paradigm is in place, an age of "normal science", dominated by disciplined puzzle-solving in agreed-upon limits, replaces radical innovation. Kuhn also stressed that paradigms are social as well as conceptual systems. They help define professional legitimacy by deciding what research can be funded, which findings are publishable, and who qualifies as the "expert of the trade."

In this respect, geroscience's rise mirrors Kuhn's model: it harmonized higher epistemic coherence with institutional feasibility and biomedical agendas around cellular and molecular mechanisms (see Section 2.1). Seen from this standpoint, the primary paradigm in geroscience is the damage-and-maintenance perspective, in which aging is considered the result of a series of failures of repair, regulation, and homeostasis across biological levels. The Hallmarks aren'ta new ontology. Instead, they arose within a process of consolidation of the paradigm, a quest to formalize, systematize and give shape to an established consensus.

From this viewpoint, the bibliometric asymmetries discussed below are relevant indicators of how a research field becomes organized around particular conceptual reference points.

Chronologically, three closely related formulations can be identified: Holliday's integrative maintenance account in the 1990s, the SENS repair framework in the early 2000s, and the Hallmarks synthesis in the 2010s. The original SENS proposal (de Grey et al., 2002: 453) explicitly cited Holliday (1995) and developed similar repair-oriented reasoning. This connection is relevant here because it illustrates how early formulations of the paradigm remained conceptually aligned even when later syntheses did not acknowledge them directly. By contrast, the Hallmarks (López-Otín et al., 2013), did not cite either Holliday or SENS despite their conceptual proximity, while clearly occurring in the same paradigm and using related traditions such as Kirkwood (2005), Vijg and Campisi (2008), and Gems and Partridge (2013). This asymmetry is therefore not evidence of theoretical disagreement but of partially independent citation traditions within a shared framework.

Quantitative comparison supports this interpretation. The Hallmarks of 2013 has garnered around 18800 citations but the current 2023 edition has already surpassed 5800 citations. By comparison, the SENS program (2002) had received about 280 citations, and Holliday's integrative damage-maintenance accounts (1995, 1996, 1998), total less than 500 overall. In other words, these frameworks belong to the same intellectual context and address similar explanatory problems, yet one of them came to shape the research agenda more decisively. This suggests that paradigmatic influence does not necessarily follow from theoretical originality, but rather from the capacity of certain representations to reorganize how scientists communicate, classify problems, and structure inquiry.

Along these lines, and what will be more important in future sections, the omission of SENS by Hallmarks, version 2013 and 2023, is not an oversight. Although it shares the same underlying ontology, SENS has maintained a strong public and translational profile, often associated with ambitious radical life extension proposals. This positioning may have contributed to its weaker integration into the academic core of geroscience, thereby reinforcing the Hallmarks as the framework through which the discipline implicitly defines what counts as central research (García-Barranquero and Pérez-González, 2025). A notable shift in citations later emerged, reflecting a strategic effort by SENS to appropriate the Hallmarks' success and integrate it into its own narrative (Bakula et al., 2019; de Grey, 2013a, 2015, 2020, 2023; Lewis and de Grey, 2024). This strategy may also be interpreted as a form of academic reinforcement: since the SENS program has attracted substantial conceptual criticism (discussed later), aligning with the Hallmarks allows the program to strengthen its standing among academic peers by demonstrating mechanistic compatibility.

Altogether, the bibliometric data suggest a Kuhnian interpretation: the appearance of the Hallmarks is not the dawning of a new paradigm but the crystallization of an old paradigm. The citation structure indicates the shift from conceptual plurality to stabilization of "normal science", where a previously dispersed set of ideas finds a common exemplar in which to organize itself. From a Kuhnian perspective, such patterns are not anomalies but expected features of paradigm formation. Conceptual convergence frequently precedes the full integration of research communities and citation networks. What appears bibliographically as partial disconnection may therefore correspond epistemically to the gradual stabilization of a shared paradigm. In this sense, the success of the Hallmarks reflects not explanatory novelty but their ability to function as a stabilizing reference point capable of aligning research agendas and providing a common conceptual vocabulary (see Tartiere et al., 2024).

As Kuhn expressed (1970), exemplars are model cases—those that orient the practice of "normal science"—enlisting shared problem–solution patterns of knowledge to shape perception, experimentation, and interpretation within an accepted conceptual order. The Hallmarks fulfill this function in terms of the damage-and-maintenance paradigm in that they translate its overall logic into a communicable heuristic that unifies, aligns, and standardizes research, vocabularies, and discourse.

Building on the work of Gems and Partridge (2013), Gems' (2024) recent elaboration suggests that dissatisfaction with purely damage-based explanations has led to renewed interest in evolutionary interpretations grounded in antagonistic pleiotropy (see again, Williams, 1957). Rather than proposing strongly programmed aging, these perspectives reinterpret longstanding theoretical contributions as alternative causal accounts that emphasize the persistence of biological programs as opposed to the progressive accumulation of damage.

Moving forward, we provide a brief overview of the different approaches to the Hallmarks (Gems, 2024: 5–16). Firstly, Blagosklonny's (2006) *hyperfunction theory* interprets aging as the late-life continuation of growth-promoting pathways that become pathological once selective pressure declines. Second, de Magalhães' (2012) *developmental theory* similarly frames aging as a temporal mismatch between developmental programs optimized for early life and their persistence into later physiological contexts. Finally, Dilman's (1971) *ontogenetic theory* attributes aging primarily to regulatory and endocrine processes rather than cumulative molecular deterioration. These approaches are not strictly incompatible with the damage-and-maintenance paradigm, but they do shift the explanatory emphasis toward regulatory mechanisms rather than damage accumulation as the primary causal driver.

However, their treatment within the Hallmarks remains limited. Although hyperfunction theory is briefly acknowledged (see López-Otín et al., 2013: 1194), these perspectives are neither systematically integrated into the model's conceptual structure nor treated as alternative organizing frameworks. This is bibliometrically significant: not because such theories are absent, but because their presence remains peripheral relative to the organizing role that the Hallmarks have assumed. We can see it with data: Blagosklonny's hyperfunction formulation (2006) has accumulated approximately 509 citations, or de Magalhães's developmental account (2012) around 173. The impact and visibility of these articles remain extremely limited compared to the success of the Hallmarks, which has relegated the former to a far more peripheral position in the literature.

Looking at this through a Kuhnian lens, we see a classic move by institutional power. A dominant paradigm doesn't necessarily need to crush its rivals; more often than not, it just assimilates them. By folding potential alternatives into its own structure, the paradigm effectively blunts their radical edge—transforming what could have been a challenge into a harmless "minority report" that sits comfortably within the *status quo*. Other concepts are so thoroughly shaped by the dominant vocabulary from the start that they aren't even recognized as alternatives; they look like more data points on the existing map. Ultimately, the survival of these theories isnt' a sign of the Hallmarks' weakness—it's proof of their absolute success in turning would-be revolutionaries into mere footnotes.

Yet, this very success reveals the limits of a purely Kuhnian account. While Kuhn helps explain how the Hallmarks stabilize the field, his framework is effective at describing how theoretical commitments evolve gradually without paradigm replacement. In geroscience, the damage-and-maintenance orientation has not been overturned but progressively modified through the incorporation of new mechanisms and conceptual refinements.

To account for this type of theoretical continuity without assuming paradigm rupture, Lakatos' (1978) framework of research programmes offers a useful complement.

2.4. Stability and degeneration in Lakatosian terms

According to Lakatos (1978), the research programmes make Kuhn's approach more complex. At the center of Lakatos' framework is the *hard core*—the essential ontological and causal assumptions that underlie a scientific field. Surrounding this hard core is a *protective belt* of auxiliary hypotheses, models and classifications; these act as a buffer, shielding the central tenets from empirical anomalies without compromising the program's foundation. Following this logic, Lakatos differentiates between *progressive research programmes*—i.e., research which can generate new predictions and integrate anomalies in theoretically productive ways—and *degenerative research programmes*, with reactive a response to new findings while failing to produce genuine conceptual novelty. In this sense, Lakatos articulates Kuhnian historicism through Popper's (1959) critical rationalism, arguing that scientific progress does not result from sudden revolutions but rather from a comparative assessment with competing programmes and their ability to renew themselves without collapsing their theoretical coherence.

Viewed through this Lakatosian lens, the Hallmarks framework reveals a clear internal structure. The hard core corresponds to the concept of aging as the compounding of cellular and molecular injury, and it could be read as a formalization of the damage-and-maintenance paradigm in a strictly Kuhnian sense. This hard core has two dimensions. Conceptually, aging is positioned as a negative but perhaps treatable process in need of intervention and management (see Blasimme, 2021; Farrelly, 2010; García-Barranquero et al., 2024a). Methodologically, the field operates on the presumption that scientific knowledge is restricted to what can be quantified and mechanistically described. Indeed, this approach to quantification, as Lemoine (2020) and García-Barranquero and Bertolaso (2022) suggest, may work to reify complex biological processes into engineering issues rather than objects for study (see also, Rattan, 2020). This hard core is supported by an intricate protective belt, which translates the abstraction of "damage" into specific, observable biological mechanisms. The Hallmarks arguably represent the dominant protective belt of this paradigm. Their significance lies not in superseding previous theories, but in stabilizing and organizing the damage narrative within a more coherent conceptual, visual, and classificatory framework.

Building on this perspective, we propose a comparative analysis of the damage-and-maintenance paradigm, contrasting its pre-Hallmarks and post-Hallmarks protective belts.

1. The first group encompasses classical efforts such as Holliday (1998), while the second focuses on engineering-based models like SENS (de Grey et al., 2002), which shared the same ontology but lacked a dominant classificatory architecture.
2. These subsequent developments demonstrate a recognizable pattern of conceptual expansion rather than of theoretical substitution. For instance, The Hallmarks of Health (López-Otín and Kroemer, 2021) contains around 600–700 citations, whereas more recent iterations like the Hallmarks of Rejuvenation (Nunkoo et al., 2024), are still in the early stages of their citation trajectories. This discrepancy is expected, as these works represent incremental extensions of the same underlying conceptual framework.

This bibliometric pattern lends weight to the notion that new articles often crystallize to reinforce the previously elaborated protective belt. Consistent with this trend, related frames in the literature show the same dynamics: The Hallmarks are restructured, extended, or repackaged but remain anchored within the same ontological framework of damage-and-maintenance. Viewed in this light, the diverse protective belts do not serve to add any real theory novelty to protect and extend that very same central intuition. The development of new hallmarks might not be seen only as evidence of theoretical pluralism but as a byproduct of the structural depth of one research programme (Gems and de Magalhães, 2021). However, they have not generated the kind of explanatory novelty, predictive power, and anomaly resolution that Lakatos regarded as typical of truly progressive science.

Against this backdrop, the analysis of Gems and de Magalhães (2021) can also be found very helpful. They argue that the Hallmarks are less a source of conceptual limitations than a symptom in this field: a rhetorically robust but conceptually restrictive framework. The problem is the structural function given that the Hallmarks contribute to a mature programme by perpetuating it rather than reformulating it. The result is therefore a widespread consensus that is broad at best and epistemically narrow at worst. In this light, then, the question of progress is inevitable. The scientific community is not unaware of this challenge (Gladyshev et al., 2024; Talay et al., 2025). If the Hallmarks did, in fact, help to build up the cohesion of the field, it might also be asked whether they, at the same time, constrained its innovative possibilities. It is necessary to examine: (1) if this structure has been a generative scaffold in the generation of research; or (2) if conceptually it has also been a bottleneck that has curtailed theoretical imagination and methodological creativity.

To investigate this further, we may employ the philosophical systems developed by Laudan (1978), Kitcher (1993) and Galison (1997), with their corresponding normative criteria for assessing the epistemic productivity of a scientific structure—i.e. as creating and solving problems, uniting research issues, and facilitating methodological innovations. From such a view, we can consider whether or not the Hallmarks are currently generating meaningful contributions to geroscience or if they reflect the limits of an overly stabilized "normal science".

3. Problems, unification, and progress

3.1. Problem-solving and conceptual limits

Inspired by Kuhn and Lakatos, Laudan (1978) argued that progress in science should not be understood as a steady march toward "the Truth" but rather that we should measure progress by how effectively *a research tradition* evolves to handle the challenges it faces. The research tradition itself is a relatively stable constellation of theoretical assumptions, methodological commitments, and conceptual orientations that dictate what constitutes a legitimate scientific problem, what methods are acceptable, and what constitutes a satisfactory solution within a scientific community. The most classical dimension of scientific inquiry corresponds to empirical problems.[3] They emerge when scientists are working to explain enigmatic observations or appreciate features of the natural world that remain poorly understood. According to Laudan, as he relates to the historical development in scientific practice, there are three types of empirical problems at least in his view: unsolved problems, solved problems, and anomalous problems.

1. *Unsolved empirical problems* are the problems for which there is no satisfactory explanation. In certain situations, those kinds of problems are not even true scientific problems until the theory gets over them. Historical cases would be, say, the explanation of Brownian motion in nineteenth-century physics, or the flatness problem in twentieth-century cosmology. These issues demonstrate how scientific inquiry is usually further refined by making unexplained phenomena into theoretical research questions with the help of theoretical tools.
2. So*lved empirical problems*, on the other hand, are those that a theory can explain, at least adequately, by the scientific standards of its time. Such answers need not be perfect or conclusive but merely appropriate in the available epistemic circumstance. Examples that appear in the literature are the kinetic theory of gases, which was a satisfactory, albeit later clarified, description of the relationship between pressure and volume.
3. Finally, *anomalous empirical problems* occur when a rival theory does sufficiently explain a phenomenon that the dominant theory does not do or does not explain well. Then a problem is also anomalous for a theory T when a rival theory T' can explain it while T cannot, either because T provides a wrong prediction or because it doesn't explain the phenomenon in the first place. One of the most well-known examples is Mercury's perihelion precession, a phenomenon that was not satisfactorily explained by Newtonian gravitational theory but that Einstein's general theory of relativity was able to explain. Such anomalies often play a crucial role in scientific change because they expose the comparative limitations of competing research traditions.

One of the outstanding unsolved empirical problems in geroscience, however, is not that of Hallmarks itself but of the broader translational project of the field. Some of these potential goals are to be able to delay aging to a great extent, reverse biological deterioration, and carry out more vigorous rejuvenation interventions (de Grey and Rae, 2007; Sinclair and LaPlante, 2019). Notwithstanding the great strides in the understanding of molecular and cellular factors implicated in aging, no method has been shown so far to exert a sustained rejuvenation influence on humans. The structural tension between mechanistic knowledge and therapeutic efficacy has been exacerbated by a proliferation of anti-aging narratives, which, as noted by Gems et al. (2024) and Aparicio (2025), has widened the gap between scientific reality and public support. This discrepancy threatens to discredit geroscience itself if the available clinical data do not support the ambitious statement made about the modifiability of aging.

[3] See again, DiFrisco and Orzack, (2026): 1.

Concurrently, the Hallmarks framework has evidently helped to solve many empirical issues according to modern science standards. In Laudanian terms, these would qualify as solved problems. Its main contributions include the standardization of the conceptual vocabulary of cellular aging, the merging of different molecular mechanisms into a common explanatory structure, and the enhancement of experiment comparability across laboratories (López-Otín et al., 2013; López-Otín et al., 2023). Moreover, targeted interventions to single hallmarks—like mTOR modulation or senolytic approaches—have led to quantified benefits for experimental systems (Campisi et al., 2019). These points are not intended as a final answer to the problem of aging; rather, they serve to illustrate the empirical utility of the framework and the limits of its methodology.

However, there are still some empirical anomalies. These correspond to observations that do not easily fit into the dominant explanatory model. For instance, the observed partial reversibility of biological age following physiological stress or medical intervention (Poganik and Gladyshev, 2023) challenges strictly cumulative interpretations of aging.

Beyond their empirical performance, Laudan emphasizes that research traditions must also be evaluated according to their conceptual problems. Conceptual problems do not derive directly from empirical failures, but rather from inadequacies in the internal infrastructure of a theoretical framework, or from tensions in the theoretical formulation with alternative approaches. Laudan distinguishes between these in this sense:

1. *Internal conceptual problems* arise when a theory suffers from logical inconsistencies, conceptual fragmentation, circular definitions, or a general lack of coherence. In the past, the suspicion surrounding Newtonian mechanics comes from its development period as one such classic historical example is that many years later Newton's use of "action at a distance" is the notion used in his gravitational theory, because these notions seemed overly conceptually problematic and reminiscent of earlier metaphysical explanations. When the physical and/or metaphysics of different theories are compared, the resulting external conceptual problems come into play. When a theory is incompatible with another (as Copernican astronomy and parts of Aristotelian physics conflict), where the acceptance of one theory diminishes the plausibility of another (as with Cartesian physiology and Newtonian mechanics), or when the explanatory link between theories does not seem to be explained by what we would expect from other theories.

The first array of conceptual problems has to do with the definition of aging itself. The Hallmarks paradigm describes aging "as a progressive loss of physiological integrity leading to functional decline and increased vulnerability to death" (López-Otín et al., 2013: 1194). This definition presents aging primarily as a process of biological deterioration associated with damage accumulation. However, this characterization is not alone nor universally accepted in the field of aging biology. As Lemoine (2020) has shown, a number of criteria have been used to define aging, including structural damage, functional decline, phenotypic change, depletion of biological reserves, or increased mortality risk. Indeed, Golubev (2021) has pointed out that while the Hallmarks framework may function nominally, as an inventory of correlated traits rather than as a full-fledged causal theory. It indicates that the framework mainly accounts for the manifestations of aging rather than the root causal architecture. Similarly, de Magalhães (2024) has pointed to the ongoing confusion over what are referred to as *drivers* and *passengers* of aging, concluding that some hallmarks are downstream rather than primary causal pathways. In summary, these critiques imply that the framework's capacity to arrange observations may outstrip its competence to account for them.

A second, internal conceptual problems relate to the explanatory architecture of the model. Even though these factors make a strong organizational taxonomy, such as primary, antagonistic, and integrative hallmarks, there are no strong mechanisms for establishing causal ordering among these processes, and none of them account that can fully explain why aging becomes a systemic biological phenomenon. As Gems and de Magalhães (2021) maintain, the efficacy of the model is not so much in coherence as in heuristic (though not in coherent explanation) and organizational (but not a cohesive explanatory one).

A further intrinsic conceptual issue is around the danger of circularity in the conceptualization. Numerous biological processes are categorized as hallmarks due to their association with age-related changes or variance across lifespan, which are then employed as proof for their causal contribution to aging. As Keshavarz et al. (2023) argue, this type of reasoning makes it difficult to distinguish the primary causes of aging from their downstream consequences or effects that are merely contingent on the biological context.

The Hallmarks framework has seldom been systematically compared to other models in the literature. We evaluate these various frameworks by distinguishing between those that are comparable, those that offer an alternative, and those that represent potentially competing approaches.

Comparable perspectives include other damage-oriented frameworks, such as the SENS program (de Grey et al., 2002), which operate within the same conceptual tradition. These frameworks may have fundamental ontological parallels to the Hallmarks framework, although the most distinctive difference is the conceptual organization or strategic orientation.

While alternative approaches may offer theoretical lenses that add additional explanatory emphases, they do not necessarily reject the damage paradigm. These theories include hyperfunction (Blagosklonny. 2006), developmental (de Magalhães, 2012), and ontogenetic (Dilman, 1971).

Deeper still are those frameworks that genuinely rival the prevailing model, offering the potential to substitute the Hallmarks as the dominant research tradition. While some ecological and evolutionary theorists have questioned the centrality of the "damage account" as a universal explanatory framework, no comprehensive rival programs have yet emerged that can explicitly claim a superior understanding of geroscience's core topics. This indicates that while the field has some theoretical diversity, it lacks full-fledged pluralism in the Laudanian sense (cf. Guillermain, 2022; see also this classic analysis, Feyerabend, 1975). The absence of a robust competitor may underscore the necessity to further investigate the structuring role of the Hallmarks (Kaeberlein, 2025). It raises the question of whether the prevailing strategic uncertainty—or even relative stagnation—in the field may be attributed to a heavy dependence on a conceptual system that, in practice, serves to monopolize the problem space for the discipline (Cohen et al., 2020a; Gladyshev, 2024). Instead of a battle between competing research traditions, geroscience appears to have coalesced into a stabilized problem-space centered on the damage paradigm. While such stabilization may lead to increased empirical productivity, it can also create constraints in the comparative assessment of rival traditions in the area of problem-solving capacity.

From the Laudanian perspective, despite its empirical efficiency, the Hallmarks programme is not without "conceptual stasis". In other words, its conceptual framework suffers from unresolved ontological and causal issues, as well as an engagement with competing paradigms that remains insufficiently broad.

3.2. *Context and authority in scientific success*

Another way of understanding scientific progress is through unification. As Kitcher argues (1993), influential frameworks do not merely account for isolated phenomena; they reorganize entire fields by bringing disparate findings under a common conceptual scheme. Their value lies not only in description, but in their ability to structure questions, methods, and research agendas within a shared logic.

This integrative ambition is especially visible in one of geroscience's foundational questions: what aging is. Recent international surveys show that disagreement over the nature, onset, and reversibility of aging, among others, remains one of the field's central conceptual problems (Cohen et al., 2020a; Gladyshev, 2024). Within this landscape, the Hallmarks framework can be understood as one of the most influential attempts to provide a unified foundation for contemporary geroscience.[4] As discussed above, it clearly operates within the damage-and-maintenance paradigm and translates that broader orientation into an especially successful classificatory map. Its importance lies in having provided a common language capable of organizing a wide range of biological processes within a single interpretive schema.

This is where Kitcher's point becomes especially useful. Unification does not succeed only because of theoretical elegance. Frameworks also survive because they align with the wider aims and dynamics of scientific practice. In geroscience, the Hallmarks have moved especially well within those extra-theoretical dimensions: they offer a stable vocabulary, a pedagogically effective schema, and a platform for coordinating data, disciplines, and expectations. Their authority, in other words, cannot be explained only by what they say about aging, but also by how effectively they organize the field that studies it.

The unificatory success of the Hallmarks becomes even more visible when contrasted with comparable integrative proposals that never achieved the same degree of institutional permanence. This becomes clearer when the Hallmarks are compared with a conceptually proximate model such as SENS. A brief reconstruction of the literature shows that SENS has attracted a large number of explicit scientific and philosophical criticisms. While these replies take different forms, many of them can be grouped around a small number of recurring objections, such as concerns about its empirical grounding, its engineering-oriented framing of aging, and its theoretical assumptions. A closer look at the citation record of key SENS publications further suggests that the secondary literature is predominantly critical, engaging with the program as a point of contention rather than a supportive foundation (Section 2.3).

For instance, Olshansky, Hayflick, and Carnes (2002) argue that it functioned less as science than as a form of biotechnological marketing, relying more on rhetorical persuasion than on robust empirical support. Gems (2011) describes it as a "wish list of interventions," suggesting that it substituted engineering optimism for biological understanding. de Magalhães (2014) emphasizes its theoretical vacuity and its neglect of evolutionary explanations of aging. Next, (Rattan, 2020) argues that SENS exemplifies a broader tendency in geroscience to recast complex biological realities as mere engineering problems. García-Barranquero and Bertolaso (2022), moreover, identifies internal tensions in SENS around repair, rejuvenation, and its machine-like conception of the organism. These examples are merely representative; a full reconstruction of the critical literature would exceed the scope of this article.

Taken together, these critiques suggest that SENS has functioned not only as a research proposal but also as a focal point through which implicit standards of scientific legitimacy within geroscience have been articulated and negotiated. In this sense, the controversy surrounding SENS may also be understood as part of the field's ongoing process of boundary-work, through which distinctions between speculative engineering visions and empirically grounded biological research are continuously drawn and redrawn Ultimately, this is reflected in the literature itself: the amount of intellectual friction generated by SENS—largely to contest its premises—far exceeds the critical interrogation directed at the Hallmarks.

Paradoxically, many of these same criticisms—overreliance on damage as a singular cause, conceptual circularity, and the reduction of aging to a set of measurable processes—could be applied to the Hallmarks as well. The key difference lies not in their ontology, which is largely shared, but in their contexts of legitimation. SENS emerged on the margins of institutional science, linked to the Methuselah Foundation and to de Grey's public persona. In contrast, the Hallmarks were born at the very center of academic biomedicine: a review article published in *Cell* in 2013 by Carlos López-Otín, Maria Blasco, Linda Partridge, Manuel Serrano, and Guido Kroemer, and presented in a visually compelling format that condensed the complex biology of aging into a single iconic image. The 2023 *Cell* update expanded that taxonomy while reaffirming its canonical status.

The question, then, is not only why the Hallmarks have been so successful, but also why they have been criticized so much less than frameworks built on strikingly similar assumptions.

3.3. Instrumentation and coordination

Another path toward scientific progress involves instrumental and procedural developments. Although the dominant image of scientific revolutions suggests that they are driven almost exclusively by entirely new and highly disruptive theoretical approaches, this perspective highlights the fundamental role of instruments and experimental techniques in the production of knowledge. Scholars associated with the well-known "Stanford School of Philosophy" have shown that the history of science has been shaped not only by conceptual discoveries but also by technical innovations that transform what can be seen, measured, and manipulated—from the microscope and X-ray diffraction to nuclear magnetic resonance, particle accelerators, and the use of artificial intelligence in molecular biology. This type of transformation has been described as a Galisonian revolution (Galison, 1997; see also, Franklin, 2007).

From that viewpoint, a second dimension, particularly relevant for our analysis, emphasizes the organizational role of certain conceptual frameworks as epistemic infrastructures capable of coordinating diverse scientific communities. From this approach, progress can also be understood as the result of the capacity to generate practical intelligibility among different research cultures—namely theorists, experimentalists, and instrument builders—even in the absence of full theoretical consensus. In this sense, progress does not necessarily depend on conceptual unification, but rather on the possibility of effective cooperation through what Galison calls trading zones: spaces where practical coordination can be achieved without complete theoretical integration.

It is precisely from this angle that we propose to situate the role of the Hallmarks. Rather than constituting a theory of aging or a Kuhnian paradigm shift, their relevance may instead be understood as a form of organizational progress, grounded in their ability to coordinate scientific practices, stabilize experimental languages, and structure research agendas. From this perspective, the Galisonian framework may explain the success of the Hallmarks better than much of the existing literature, which tends to evaluate their influence primarily in biological terms rather than as a form of organizational infrastructure for the field. The Hallmarks can thus be understood as a stabilized trading zone: a space in which diverse traditions—molecular biology, pharmacology, genetics, and biotechnology—converge without the need to share a single theoretical framework. This mediation enables practical communication and methodological compatibility.

However, this same organizational success may also entail epistemic costs. The consolidation of the Hallmarks has contributed to reinforcing

[4] In response to this question, at least two limiting positions have emerged. One is associated with Cohen et al. (2020b), who suggest that aging may not correspond to a single biological process at all, but rather to a heterogeneous set of partially independent phenomena grouped under one label. The other is de Grey's interventionist response (2013b), according to which the urge to define aging may be less important than the practical task of repairing or modifying its manifestations. These two views highlight the complexity of the issue, philosophically speaking.

a form of technological solutionism in which understanding a phenomenon becomes progressively equivalent to identifying possible interventions upon it. In this context, contemporary data-driven biology tends to privilege what can be measured or manipulated (Leonelli, 2019) over more fundamental questions about the nature of aging. What was initially proposed as a heuristic scheme to organize aging mechanisms appears to have gradually transformed into an implicit ontology of aging itself, much like SENS. If geroscience is "at a precipice" (Poganik and Gladyshev, 2023: 1), it implies not only an opportunity but also a conceptual risk. So, while we are witnessing the rapid growth of the field of intervention-oriented research and the growing expectation of significant translational breakthroughs (Gems et al., 2024), it may also signal a technological turning point for the field. But this same momentum also calls for questioning what's an increasingly visible form of technological solutionism in the view that knowledge about aging will emerge naturally through the accumulation of interventions, datasets, or therapeutic platforms. Understanding aging cannot be reduced to the hope that sufficiently advanced technologies will ultimately deliver answers (García-Barranquero and Bertolaso, 2022). That risk, if not the problem, would be that the quest for interventions will start to replace, among other things, the harder work in establishing what aging is, why it occurs, and how we should measure success in geroscience.

In this sense, the stabilization of a common language not only facilitates scientific cooperation but also delimits the space of questions considered legitimate. That which cannot be formulated within the vocabulary of damage, dysfunction, and repair risks becoming epistemically marginal. This may help explain why the scientific community has not extensively questioned this framework: it fits comfortably within assumptions that have long remained largely unquestioned (see also, Kuhn, 1970). The success of the Hallmarks may therefore be understood not only as a device for scientific coordination, but also as a cognitive filter that helps define which types of explanations appear plausible and which problems deserve scientific attention.

At this point, we read the proposal of Okholm (2024) in order to achieve our goal. He suggests understanding geroscience as a boundary object: a structure that enables collaboration among different scientific communities while simultaneously reproducing particular epistemic hierarchies. Following this line of thought, we propose interpreting the Hallmarks not simply as a biological framework, but as an epistemic infrastructure that simultaneously facilitates scientific coordination while stabilizing a particular way of thinking about aging, privileging mechanistic and interventionist approaches over alternative perspectives.

All of the above can be empirically supported through recent bibliometric analyses (Perez-Maletzki and Sanz-Ros, 2025). Computational studies of the aging literature have identified approximately thirty dominant thematic categories structuring contemporary research, showing that mechanisms associated with the Hallmarks—such as cellular senescence, oxidative stress, and mitochondrial dysfunction—function as stable conceptual clusters within the scientific literature. These findings suggest an increasing thematic convergence, reinforcing the idea that the Hallmarks function not only as an explanatory scheme but also as an infrastructure that organizes the very practice of contemporary geroscience.

4. Conclusions

In this work, we have examined in detail what is probably the most influential conceptual framework in contemporary geroscience, analyzing the Hallmarks through philosophical tools that had also already been partially invoked in the literature of the field itself (Cohen et al., 2020a; Gems and de Magalhães, 2021). Rather than introducing external perspectives, our objective has been to systematically develop intuitions already present in the debate, using different approaches to better understand the different functions that the Hallmarks play within the discipline. Philosophy, Godfrey-Smith (2014: p. 4) reminds us, is not above science, but can allow the distance to perceive how components of a field fit together and how a field's wider intellectual structure develops.

In Kuhnian terms, the dominant model in contemporary geroscience understands aging as the accumulation of molecular and cellular damage, and the Hallmarks represent its most characteristic exemplar: the canonical structure through which normal science operates, defining what counts as valid evidence, what types of intervention are considered pertinent, and what problems are considered solvable. From a Lakatosian perspective, this same process can be interpreted as the consolidation of a theoretical core based on the same ontology, surrounded by a protective belt of alternative models and classifications —such as SENS— in the debate, and which allow the empirical horizon to be expanded without questioning their fundamental assumptions. From Laudan's approach, this process can be understood as a reorganization of the problem-space of the field, where the consolidation of the framework has not eliminated the open empirical or conceptual problems, but has restructured them within a more coherent methodological environment. From Kitcher's view, its success can be explained in more contextual than purely theoretical terms, as a result of its organizational capacity and its diffusion as a common reference. Finally, following Galison's ideas, the Hallmarks can be understood as a pragmatic device that harmonizes methods, data, and scientific communities, functioning as a stabilized trading zone in which collaboration can continue even in the absence of full theoretical convergence.

This reading also connects with a concern that is increasingly visible in the recent debate: the problem of consensus in the biology of aging (Cohen et al., 2020a; Gladyshev, 2024). Faced with the persistence of disagreements on fundamental questions, our analysis suggests that the role of the Hallmarks in this context may have been greater than has usually been recognized. Rather than resolving the disagreement, they have contributed to structuring the field by allowing its practical coordination despite the absence of theoretical consensus, and we believe we have offered sufficient reasons to justify this interpretation. This structuring function becomes especially clear when one considers the recent catalogue of 100 open questions in research in the field (Talay et al., 2025). What is significant is not only the magnitude of the issues identified, but the fact that many of them are formulated within the conceptual space that the Hallmarks have helped to stabilize. Returning to Laudan, this can be interpreted as evidence of how a framework contributes to defining which problems are considered scientifically relevant. In this sense, an additional question could be formulated explicitly: whether, alongside the problems identified, one should also directly ask about the conceptual framework that organizes them, as has already been suggested in the recent debate (see also, Kaeberlein, 2025).

In this same context, this reflection naturally connects with another emerging issue in the recent debate that refers not so much to mechanisms as to the goals of the field. What exactly should geroscience attempt to achieve? Compression of morbidity, improving health span, radical life extension, or even rejuvenation? (Aparicio 2025; Le Bourg, 2022; Rattan, 2024). The persistence of these questions suggests that uncertainty about the aims of the discipline may be as important as uncertainty about aging itself. Precisely for this reason, reflecting on the objectives of the field also implies examining the conceptual frameworks that guide those decisions and allows us to reconsider our own position within it: why we use the Hallmarks as an organizing framework, what type of scientific programme they favor, and to what extent their adoption responds exclusively to epistemological reasons or also to structural dynamics of the disciplinary development itself, to cite some research lines derived from our study.

For this reason, the purpose of this work has not been to question the Hallmarks' empirical value, but rather to situate them within a broader reflection on how scientific progress in geroscience is constructed and on the role that conceptual frameworks play in the future orientation of the field.

Funding

Pablo García-Barranquero was supported by the Spanish Ministry of Science and Innovation (Project PID2023–149342NB-I00). Victor J. Luque was supported by the Spanish Ministry of Science and Innovation (Project PID2024–155508NA-I00). Both authors were supported by the Spanish Ministry of Science and Innovation (Project PID2021–128835NB-I00) and by the Conselleria d'Innovació, Universitats, Ciència i Societat Digital of the Generalitat Valenciana (Project CIGE/2023/16). Funding for open acces charge: Universidad de Málaga / CBUA.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We are grateful for the comments and suggestions provided by Marta Bertolaso, João Pedro de Magalhães, Antonio Diéguez, David Gems, Gregor Greslehner, Valeriano Iranzo, Guillermo López-Lluch, Sául Pérez-González, and Cristian Saborido, which substantially improved our article. We also extend our thanks to the audiences of the *Philosophy of Medicine: Approaches and Perspectives* conference in Sofia, Bulgaria (2024) and the *Second Workshop on Causation, Mechanisms, and Evolutionary Biology* in Valencia, Spain (2025), where we presented early versions of our work and received valuable feedback. Finally, we would like to acknowledge and sincerely thank Mara Alexa for her invaluable work in English-language revision of this manuscript.

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